

UNIT 3

BRANCHES OF ELECTRICAL ENGINEERING

What is Power Engineering

Power Engineering is focused on the design of large-scale systems for the generation, transmission, and distribution of electricity. Electrical power is an energy source created from electrical charge and Power Engineering focuses on these large-scale applications involving extremely high current and voltage.

The process of generating, transferring, and distributing large amounts of power is a necessity for modern society, yet difficult and so Power Engineers provide solutions to these difficulties. Conversion between DC and AC power is a requirement when distributing electricity to residential areas. Some Power Engineers work on generating the power that enters the electricity grid, while others design transformers, generators, and motors that utilise or modify this power.

Power Engineering graduates now have the opportunity to work in the emerging area of renewable energy sources and modern power technology. Careers are available across many large companies such as regulatory agencies, utility companies, and consulting agencies. Power Engineers often split their time between on-site work at power plants, substations, and in labs conducting research.

What is Control Engineering

A Control Engineer designs ways to control the behaviour of dynamic systems. Any system that constantly changes is considered dynamic. Some common examples of dynamic systems are a car that is travelling on a road, the temperature outside, and manufacturing equipment.

Control Engineering aims to create stability in these systems through various strategies. The first step to control a system's behaviour is to measure and identify the system. Depending on the application of the control system, a Control Engineer has access to a wide range of sensors which can be used by other Engineers to measure any variable(s) to be controlled (e.g. pressure, voltage, acceleration, and force).

For complex systems, which is usually the case in real-world applications, computer simulations based on first principles (mathematical equations) are used to investigate how to best control these variables. A Control Engineer will perform calculations and design software to best control the system as required. They also design the physical units that execute the software (microcontrollers). Students of Control Engineering tend to be highly theoretical, and are required

to be proficient with software development. Control Engineers are employed in the automotive, aerospace, and manufacturing industries. Most work is performed on a computer in an office environment, however, there can also be fieldwork to measure and test new control systems.

What is Electronics Engineering

Electronics Engineering is involved with the development of small-scale electronic systems to transmit information. All electronic devices are powered by a battery or mains electricity source. Unlike the other disciplines within Electrical Engineering, Electronics Engineering deals exclusively with active electrical components. Active components of a circuit are designed by Electrical Engineers to add power (for example amplifiers).

The fundamental skill of an Electronics Engineer is their understanding of the flow of electricity (of electrons) in a circuit, enabling them to design circuit boards. They also design and develop device schematics and ensure the compatibility of their devices with other electronic systems. A large quantity of their work includes testing components for quality and creating prototypes for clients.

With some experience, Electronics Engineers often move into project management, however many also work in product development and research. They are more than capable of starting their own consulting business or designing their own products. The majority of an Electronics Engineer's time is spent in an office or laboratory working but can include occasional factory visits.

What is Computer Engineering

Computer Engineering is dedicated to the careful integration of software and hardware in computers. At its most basic, a computer is essentially an engine that performs computations (e.g. solving mathematical functions to calculate and organise information from data). A Computer Engineer aims to create a harmonious relationship between the hardware and software to enable computers to work as designed.

A Computer Engineer can be involved in the design of sensors, circuit boards, and even operating systems. Many of these are incredibly large-scale projects which require Computer Engineers to work in teams and to collaborate with Software and Electronics Engineers. Computer Engineers spend the majority of their time dedicated to a strategic role to evaluate the trade-offs between hardware and software.

Computer Engineering graduates also possess business and management skills to assist in the delivery of effective computer systems at the lowest possible cost or to meet a client's needs. The main industries Computer Engineers are employed in include video games, defence, and consumer electronics, and

various branches of information technology. Today, the widespread use of computer systems provides a multitude of career paths for Computer Engineers.

What is Software Engineering

The design and development of software fall within the domain of Software Engineering. Software is, fundamentally, an organised script that sends a series of instructions to hardware to perform a particular task.

Software Engineering ensures the end product is user-friendly, cost-effective, and accurate. While computer engineering is a similar field, it focuses less on the practical application of software. Software Engineers also design algorithms and compilers used to execute programs.

After a few years of experience, Software Engineers can often be responsible for overseeing entire projects in a management role, supervising a team of programmers. Common daily tasks include setting design targets and troubleshooting existing code problems. All IT companies employ Software Engineers to design customised software for various projects requested by clients. Software Engineers may also work as freelance game and application developers.

What is Telecommunications Engineering

Telecommunications Engineering is focused on the devices and mediums used to transmit communications information in the form of words, sounds, or images over great distances. The main mediums a Telecommunications Engineer develops systems and solutions within are telephone, radio, television, and the internet.

Project planning, supervising and organising, and overseeing the installation of these devices are tasks that Telecommunications Engineers may perform as part of their daily duties. One of the most important skills that any good Telecommunications Engineer must have is in both digital and analog signal conditioning and processing. This is because all telecommunications are eventually (if not already) converted into electrical signals which can be modified, improved, and enhanced.

Graduates of Telecommunications Engineering can specialise in any one of the following areas: GPS, internet, mobile networks, and radio transmission. Telecommunications is a research area that is technically challenging and offers well-paid contracts to those who excel at it. Telecommunication companies always seek out the services of engineers to develop their network base.

Exercises

1. Answer the following questions.

- a) What is in the focus of power engineering?
- b) Give some examples of work and duties that a power engineer might work on.
- c) What are the possible career options for a power engineering graduate?
- d) What does control engineering deal with?
- e) Give the definition and examples of a dynamic system.
- f) Give some examples of work and duties of a control engineer.
- g) What is the role of sensors in control engineering?
- h) What is the role of computer simulations in control engineering?
- i) What are the possible employment options for a control engineer?
- j) What is electronics engineering involved with?
- k) What are active components of a circuit?
- l) What are the required skills of an electronics engineer?
- m) What are the possible career options for electronics engineers?
- n) What is computer engineering dedicated to?
- o) Define a computer.
- p) What is the aim of a computer engineer?
- q) What can a computer engineer be also involved in?
- r) What are the main industries the computer engineers are employed in?
- s) What falls within the domain of software engineering?
- t) Define software.
- u) What is the difference between software engineering and computer engineering?
- v) What are the possible career options for software engineers?
- w) What is telecommunications engineering focused on?
- x) What are the main media within which a telecommunications engineer develops systems and solutions?
- y) What duties in relation to the media a telecommunications engineer may perform?
- z) What is one of the most important skills for a telecommunications engineer?
- aa) What are the fields in which the graduates of telecommunications engineering can specialise?

UNIT 3

2. Match the type of engineering with the responsibilities and career options.

Power Engineering	creation of stability in dynamic systems	project management, product development and research, consulting business, product design
Control Engineering	Development of systems and solutions for telephone, radio, television and the internet	overseeing of projects in a management role, setting design targets, troubleshooting existing code problems. design of customised software, freelance game and application developers.
Electronics Engineering	integration of software and hardware, design of sensors, circuit boards and operating systems	GPS, internet, mobile networks, radio transmission
Computer Engineering	generating of power, design of transformers, generators and motors	automotive, aerospace and manufacturing industries
Software Engineering	design of circuit boards, design and development of device schematics, ensuring of compatibility between the devices and electronic systems, testing of components for quality, creating of prototypes	video games, defence and consumer electronics, various branches of information technology
Telecommunications Engineering	design and development of software	renewable energy sources, modern power technology, regulatory agencies, utility companies, consulting agencies